A paleomagnetic and K–Ar study of the Umfraville Gabbro, Ontario

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Two anti-parallel remanence directions have been isolated from samples of the Umfraville Gabbro situated in the southern part of the Grenville Province, Ontario. The magnetization yields a pole position at 160°E, 7½°S, A_{95} = 9.0°. Because of the dispersion of remanence directions, however, individual site poles plot over much of the Grenville Track. Regardless of direction and polarity of remanence, all samples unblock within a narrow temperature range of 525–560°C. It is inferred that the magnetization process has involved slow cooling with both TRM and TCRM components being generated. Although one sample of zircon extracted from a genetically related syenite probably provides an estimate of the time of emplacement of 1160 Ma, the magnetization may be more closely related to the K–Ar age. Whole rock K–Ar ages range from 873 Ma to 1130 Ma, but all data points lie close to a 911 Ma isochron, with an initial $^{40}Ar/^{36}Ar$ ratio of 805. The isochron age is in close agreement with ages of ^{36}Ar-free samples.

On a isolé deux directions de rémanence anti-parallèles dans des échantillons de gabbro d'Umfraville provenant de la partie sud de la province de Grenville. L'aimantation donne une position de pôle à 160°E, 7½°S, A_{95} = 9.0°. Toutefois, à cause de la dispersion des directions de rémanence, les pôles des sites individuels s'étalent sur la plus grande partie de la trace du Grenville. Si on néglige la direction et la polarité de la rémanence, tous les échantillons débloquent dans un intervalle assez court de température entre 525 et 560°C. On émet l'hypothèse que les processus d'aimantation résultent d'un lent refroidissement avec génération des composantes TRM et TCRM. Bien qu'un échantillon de zircon extrait d'une syénite génétiquement apparentée fournisse une estimation probable du temps de la mise en place de 1160 Ma, l'aimantation pourrait se rattacher davantage à l'âge au K–Ar. Les âges au K–Ar de roche totale vont de 873 à 1130 Ma, mais tous les points reposent près de l'isochrone de 911 Ma avec un rapport initial $^{40}Ar/^{36}Ar$ de 805. L'âge d'isochrone est très compatible avec les âges des échantillons ne contenant pas de ^{36}Ar.

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Introduction

The Umfraville Gabbro, located about 20 km south of Bancroft, Ontario, is intruded into the metasediments and metavolcanics of the Hermon and Mayo Groups. In contrast to many of the other gabbroic bodies in the Bancroft–Madoc area, the Umfraville intrusion exhibits very little internal metamorphism and is believed to have been intruded rather late in the complex sequence of magmatic and metamorphic pulses that have affected this portion of the Grenville Province (Lumbers 1967). The Umfraville Gabbro was studied paleomagnetically 20 years ago by Hood (1958) and that work represents one of the earliest paleomagnetic studies of Precambrian rocks. At that time cleaning methods were in their infancy and the early work of Hood involved only the partial AF demagnetization of a few of the more divergent specimens from his collection of 12 independently oriented samples. With the availability of efficient AF and thermal demagnetization equipment we undertook to re-examine this body to see if additional magnetic history could be extracted from it. During the review process of this paper, Symons (1978) published results of a study of the Umfraville Gabbro undertaken for the same reasons as influenced us. In addition to the paleomagnetic data we report new results on the K–Ar geochronology of the unit and thus derive additional information on its cooling history.

Geology, Sampling, and Paleomagnetic Methods

Although gabbro is the most abundant rock type of the intrusion other significant rock types present are pyroxenite, peridotite, olivine gabbro, Fe-Ti oxide-rich gabbro, anorthositic gabbro, and diorite. Spatially associated with the mafic rocks of the intrusive complex are intrusions of potassic syenite and pyroxene monzonite, the larger bodies of which are indicated on Fig. 1. Contact relationships cited by Lumbers (1969) indicate that the syenitic rocks probably were intruded before the mafic and anorthositic rocks were completely crystallized. Further evidence that the mafic and syenitic rocks are genetically related is provided by the fact that

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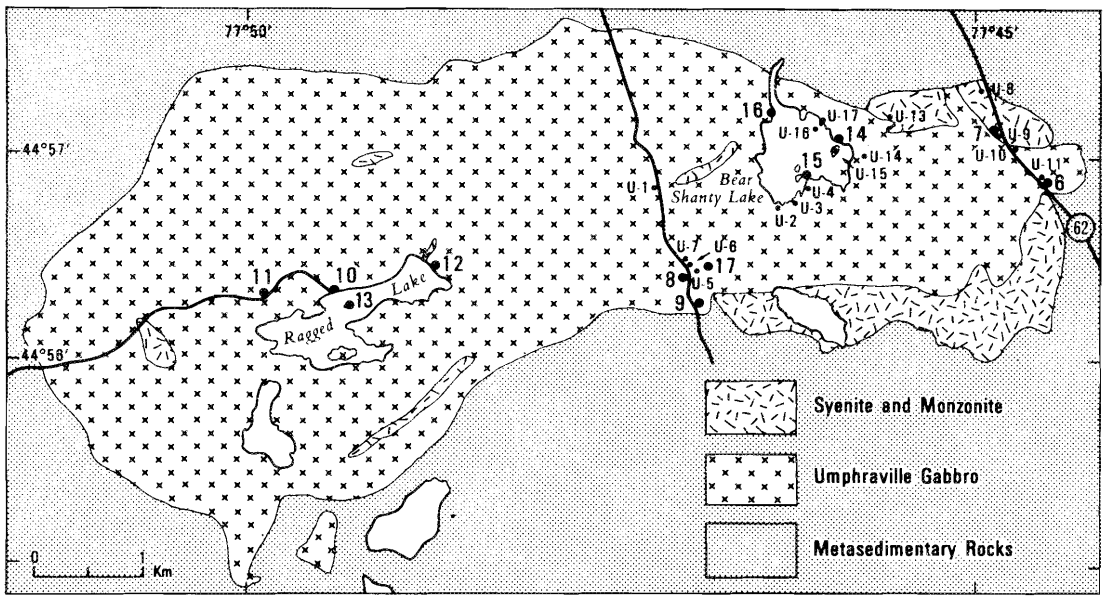


FIG. 1. Geological map of the Umfraville Gabbro compiled from maps included in the reports of Hewitt (1957, 1962) and Lumbers (1969). Field-drilled sampling sites are indicated by large numbers; location of block samples are prefixed with the letter 'U'.

the mafic rocks and some of the ultramafic rocks contain modal biotite. The gabbroic rocks of the Umfraville intrusion are distinctly richer in potassium than are the gabbroic rocks from other intrusions in the Bancroft–Madoc area such as the Thanet, Jocko Lake, and Tudor gabbros (Lumbers 1969, Table 7; Hayatsu and Palmer 1975, Table 1; this paper, Table 1). The genetic association of the syenitic rocks to the mafic rocks of the Umfraville intrusion is critical because the only information to date on the age of the body is provided by U/Pb data obtained from zircons extracted from the syenite near our site 7 (Fig. 1). Analytical results from this sample of zircon yielded a single, slightly discordant data point which when projected to 'concordia', gives a minimum age estimate of 1160 Ma (S. B. Lumbers and L. Silver, personal communication, 1978).

An early collection of block samples (U series) was made in 1963 when the field mapping of Limerick Township was in progress. At the time of collection, laboratory facilities then available to us made interpretation of the preliminary results obtained from these samples difficult. With improved laboratory equipment over the intervening years a second attempt was made at understanding the magnetic history of this body. Accordingly, during the fall of 1976 and the spring of 1977 new additional samples (SP series) were collected at 12 sites more evenly distributed over the intrusion. Generally, four independent samples per site were obtained with a portable field drill and the azimuthal component of orientation was determined with reference to the sun. Since the original block samples were generally collected from different outcrops than were the field-drilled samples, the blocks could be considered as from additional locations and so were re-drilled in 1977 and new paleomagnetic specimens were prepared from them.

Measurements of the intensity and direction of remanent magnetization were made with a Schonstedt DSM-1 spinner magnetometer. AF demagnetization was carried out in steps to a maximum of 110 mT (peak) using apparatus similar in design to that described by Roy *et al.* (1973). Step-wise thermal demagnetization, carried out on 26 specimens from the collection, was done in furnaces employing nested mu-metal shielding. One of these demagnetizers is a Schonstedt TSD-1 model; the other is a home-built model in which the samples can be slowly rotated along one axis during the cooling cycle. This rotation minimizes the acquisition of small partial thermal remanent magnetizations (PTRM's) in one plane owing to incomplete DC field cancellation. Temperature calibration experiments, temperature gradient measurements and duplicate runs in this furnace and with the Schonstedt TSD-1 indicate that blocking temperatures given are conservatively estimated to be accurate to 10°C. Polished thin sections were pre-

TABLE 1. Summary of K–Ar analytical data

Site No.	Sample	K%	$^{40}Ar_{rad}$ $\times 10^{-5}$ mL g^{-1}	^{36}Ar $\times 10^{-10}$ mL g^{-1}	$^{40}Ar/^{36}Ar$ $\times 10^{4}$	$^{40}K/^{36}Ar$ $\times 10^{5}$	Age (Ma)
6	SP-25	0.101	0.624	7.76	0.83	0.89	1130
9	SP-35A	0.458	2.032	6.93	2.96	4.52	874
9	SP-35B	0.514	2.295	5.81	3.98	6.05	873
10	SP-39*	1.163	5.302	<0.07	>694	>1041	893
10	SP-40*	2.19	9.976	0.00	(∞)	(∞)	892
12	SP-51A	0.763	3.579	2.71	13.26	19.28	913
12	SP-51B	0.730	3.432	2.74	12.54	18.20	915
15	SP-61	0.087	0.423	4.37	1.00	1.36	939
15	SP-63	0.145	0.684	3.32	2.09	2.99	917
16	SP-65	0.489	2.371	3.77	6.31	8.86	938

NOTE: *Non-magnetic fraction only.

pared from the same cores that were used for the paleomagnetic and age-dating studies. Small chips from these cores were also used for Curie point determinations, which were made employing a null method in a horizontal translation balance; at the null position the applied field used was 0.8 Tesla.

K–Ar Dating

Samples used for the potassium–argon study were selected from the field-drilled paleomagnetic samples after paleomagnetic data had been obtained from them using only the AF demagnetization technique. The selection of samples was based on the criteria of obtaining good areal coverage of the intrusion and good representation of the major rock types present in it. Unaltered or least altered samples of these major rock types were selected on the basis of thin-section examination.

The samples so selected were crushed to pass through a 20 mesh screen. Powders finer than 100 mesh were discarded. Samples SP-39 and SP-40 contained large amounts of iron oxides. For these two samples a hand magnet was used to remove magnetic grains in order to avoid interference of iron with the flame photometry determination of potassium. Each sample was then homogenized and split into two portions; one portion was loaded into an argon extraction system immediately and the other portion was further crushed to pass 100 mesh and used for potassium analysis.

The results are summarized in Table 1. Columns 5, 6, and 7 were corrected for atmospheric argon contamination of blank runs, which is 3.5×10^{-8} cm^3 STP and is reproducible to within 10%. Samples SP-35A, 35B, and SP-51A, 51B are two slices from the same cores and provide a measure of analytical precision. The ages of all samples, calculated by the conventional method, range from 873 Ma to 1130 Ma. The age for SP-25 is grossly different from the other seven ages. In the conventional interpretation, therefore, this age would be discarded as too high and the average of the remaining seven, 909 ± 25 Ma would be taken as the age of the Umfraville Gabbro.

Figure 2 shows a plot of $^{40}Ar/^{36}Ar$ *vs.* $^{40}K/^{36}Ar$ of the same data with no data being rejected. Samples SP-39 and SP-40 contained practically no ^{36}Ar and hence cannot be included in the plot nor used in the isochron calculation. For samples SP-35 and SP-51, only average values of sub-samples A and B are plotted. In this plot, all points lie close to a straight line whose slope and intercept determined by a least squares method (York 1969) yielded an isochron age of 911 ± 11 Ma and initial $^{40}Ar/^{36}Ar$ ratio of 805 ± 922. Therefore, the high conventional age for SP-25, and to a large extent, the spread among the remaining seven ages (ranging from 873 Ma to 939 Ma) are due to the effect of initial argon rather than loss of or excess of argon. The close agreement between the isochron age and the ages of ^{36}Ar-free samples (SP-39 and SP-40, Table 1) provides further support to the isochron interpretation.

The degree of discordancy in the conventionally

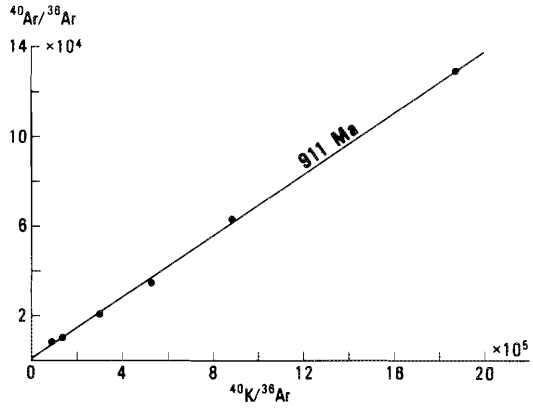


FIG. 2. Isochron plot of the whole-rock data.

calculated ages is not great because the ^{36}Ar contents of all of these samples are generally very low and the initial $^{40}Ar/^{36}Ar$ ratio is only 2.7 times the present atmospheric $^{40}Ar/^{36}Ar$ ratio. Nevertheless the Umfraville Gabbro isochron illustrates that the K–Ar isochron method can provide an unambiguous interpretation for discordant conventional ages for which the conventional method provides only qualitative and hence somewhat arbitrary interpretation. Moreover, the isochron method provides a more precise age estimate.

It may be worth noting that the ^{36}Ar-free samples contained large amounts of magnetic grains, which were removed by a hand magnet during sample preparation. The suspicion that removal of magnetic grains had some effect on the ^{36}Ar contents was rejected by a ^{36}Ar measurement on a separate slice of whole rock SP-39, which had not been subjected to the magnetic separation process. It may be interesting, however, to see if ^{36}Ar contents are uniformly low in oxide-rich rocks from other plutonic bodies.

Paleomagnetic Results

In four of the seventeen block samples (U series) and in three of the nine field-drilled sites (SP series) no satisfactory end point of acceptable within-sample/site precision could be isolated by either AF or thermal demagnetization methods. In the remaining samples/sites, one and in some cases two, characteristic remanence directions were isolated by a combination of AF and thermal demagnetization. The natural remanent magnetization (NRM) directional results of these samples and sites are shown in Fig. 3a. In Fig. 3b are shown the results from these same samples and sites after treatment. For all but two of the data points, U-12 and SP site 13, the cleaned directions plotted are those obtained at the alternating field indicated in Table 2. U-12 exhibited directional instability at AF fields greater than 40 mT. The directional trends observed in the 0–40 mT treatments were similar to those observed in the low-temperature stages of thermal demagnetization and thus the directional end point achieved by the thermal demagnetization method is used for this particular sample in Table 2 and Fig. 3b. Similarly the directional results obtained by the thermal demagnetization method are used to represent site 13; the thermal method better isolates the end point than does AF treatment alone (Fig. 4).

As is illustrated in Fig. 3b, two groupings are obtained. The most easily isolated and most widespread direction obtained from the body is one having an east-southeast (ESE) declination and a variably inclined downward direction, which we will call for convenience the B direction. A few samples from a restricted area south and west of Bear Shanty Lake exhibit a characteristic remanence having a west-northwest (WNW) declination and an upward inclination. This we will refer to as the A component. It is to be noted that the A component, however, is antiparallel to some of the more steeply inclined B directions.

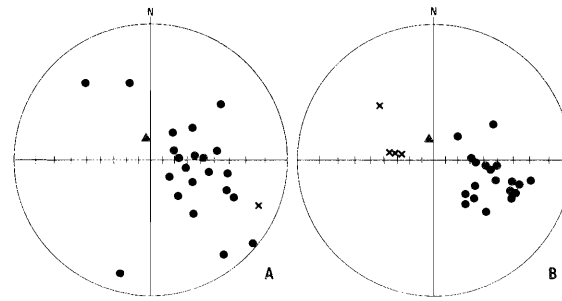


FIG. 3. NRM directions of magnetization (A) before and (B) after demagnetization treatment for all stable samples or sites. Solid circles indicate north-seeking directions on the lower hemisphere; ×'s indicate north-seeking directions on the upper hemisphere. The triangle is the present field at the collecting locality.

The B Component

Although the demagnetization techniques employed greatly reduced the directional scatter in the data (compare Fig. 3a and Fig. 3b) considerable dispersion about the mean still remains, particularly in the B grouping. The bulk of the evidence suggests that this dispersion is real and not the result of orientation errors or the failure to resolve two or more components. For many B type samples, the B component is dominant in the NRM, and AF demagnetization effects only an intensity decrease (*e.g.* SP-26-1—Fig. 5). In the bulk of the remaining samples convergence to the B direction is obtained at moderate AF fields and thereafter only an intensity decrease, without directional change, is effected (Fig. 6). Thermal demagnetization alone, or thermal demagnetization subsequent to partial AF demagnetization generally yields end points little different from those obtained from the AF treatment alone (Fig. 4). Only in one example from B component samples (as defined by the AF treatment) was there any evidence that other components are present. Figure 5 illustrates the evidence for this statement where two specimens from the same sample are compared. AF demagnetization reveals only a moderately inclined B component whereas thermal demagnetization suggests that a steep B (or reversed A) is being removed in the 530–535°C steps and that the A component

TABLE 2. Site/sample mean paleomagnetic results and summary statistics

Sample/ Site	N	NRM			Cleaned					VGP	
		Dec.	Inc.	AF/T	Dec.	Inc.	R	k	α_{95}	Long.	Lat.
*U 2	2	143	+53	30 mT	138	+52	1.999	1167.	7.3°	136°E	4°N
†U 3	2	195	+08	115 mT	278	−63	1.991	106.	24.5°	154°E	25°S
U 3	3			540-555°C	112	+68	2.950	40.	19.8°	140°E	23°S
*U 4	2	082	+38	90 mT	096	+41	1.997	298.	14.5°	171°E	12°S
†U 5	2	141	+36	90 mT	280	−54	1.998	531.	10.9°	161°E	17°S
U 5	3			530-540°C	124	+59	2.880	17.	31.2°	142°E	8°S
*U 6	2	142	+07	60 mT	135	+34	1.999	1685.	6.1°	146°E	15°N
*U 7	2	039	+60	100 mT	043	+63	1.998	456.	11.7°	179°E	60°S
*U 10	2	084	+54	110 mT	093	+56	1.999	1866.	5.8°	163°E	23°S
*U 11	2	114	+22	40 mT	117	+25	1.997	380.	12.8°	164°E	9°N
U 12	2	350	+09	40 mT	326	−07	1.985	67.			
*U 12	3	345	+30	550°C	315	−32	2.994	313.	7.0°	147°E	16°N
*U 13	2	320	+17	70 mT	124	+51	1.999	1103.	7.5°	147°E	2°S
*U 14	2	114	−08	60 mT	103	+18	1.924	13.		177°E	2°N
*U 15	2	129	+02	30 mT	107	+24	1.957	23.		172°E	3°N
*U 17	2	085	+67	70 mT	091	+62	1.998	417.	12.3°	158°E	28°S
*SP 6	4	100	+31	35 mT	106	+29	3.989	276.	5.5°	171°E	0°N
*SP 8	4	102	+43	40 mT	108	+39	3.986	216.	6.3°	164°E	3°S
*SP 9	4	112	+28	15 mT	113	+26	3.913	34.	15.9°	166°E	6°N
*SP 10	3	101	+60	35 mT	100	+44	2.982	109.	11.9°	167°E	11°S
*SP 12	4	051	+23	35 mT	059	+36	3.899	30.	17.1°	161°W	36°S
SP 13	4	050	+48	40 mT	130	+51	3.485	6.			
*SP 13	4	050	+48	540°C	145	+46	3.946	55.	12.5°	134°E	11°N
*SP 14	4	131	+69	20 mT	113	+27	3.959	73.	10.8°	166°E	5°N
*SP 16	4	118	+52	40 mT	134	+46	3.957	70.	11.0°	142°E	7°N
†SP 17a	2	066	+69	100 mT	280	−58	1.999	6141.	3.2°	158°E	20°S
SP 17a	2	066	+69	500°C	278	−55	1.999	36477.	1.3°	161°E	20°S
SP 17a	2	066	+69	550°C	097	+61	1.840	6.		156°E	24°S
*SP 17b	2	089	+47	40 mT	097	+48	1.826	6.		166°E	15°S
Mean	23				109	+44	21.530	15.	8.1°	161°E	6°S
										(dp = 7°, dm = 10°)	
Mean of site poles	23									160°E	7½°S
										(A_{95} = 9.0°)	
A component	3				279	−58	2.994	318.	6.9°	157°E	21°S
B component	20				110	+42	18.652	14.	9.0°	161°E	5°S

NOTES: N = the number of specimens in the U series and is the number of samples in the SP series. Dec. = the declination in degrees measured clockwise from North; Inc. = the inclination in degrees with north-seeking directions down taken as positive. AF/T = the cleaning treatment of alternating fields in milliTesla or of thermal treatment in °C. R, k, and α_{95} are the standard statistical parameters of Fisher (1953). VGP = virtual geomagnetic pole.
*These entries are used in the calculation of the overall mean pole.
†These entries are used in the calculation of the A component direction and pole.

itself is being unblocked at the 540°C step. Immediately prior and just subsequent to the large intensity drop (545–560°C) reversed A (or steep B) directions are obtained. Not much significance would be attached to this type of directional change close to and at the blocking temperature where it is observed in one sample from a paleomagnetic collection. However, the isolation of the reversed A direction is characteristic of most of the samples in which the A direction itself is first isolated; this will be illustrated in the next section.

The A Component

No samples exhibit the A component in the virgin state; this component has been isolated as an end point by AF demagnetization in 2 of the four cores from site 17 and in the block samples: U-3 and U-5. Thermal demagnetization isolates a similar direction in U-12, but in this sample the inclination is substantially shallower than in the other samples that exhibit the A component. Figures 7 and 8 illustrate the directional and intensity changes in two specimens from site 17. Initial remanence directions appear to be resultants of a present field component and the B direction. In AF fields of 10 mT to 80 mT the directions migrate through 150°–160° of arc on great circle paths to the A direction. Once isolated in fields of 90 mT to 110 mT, the A compo-

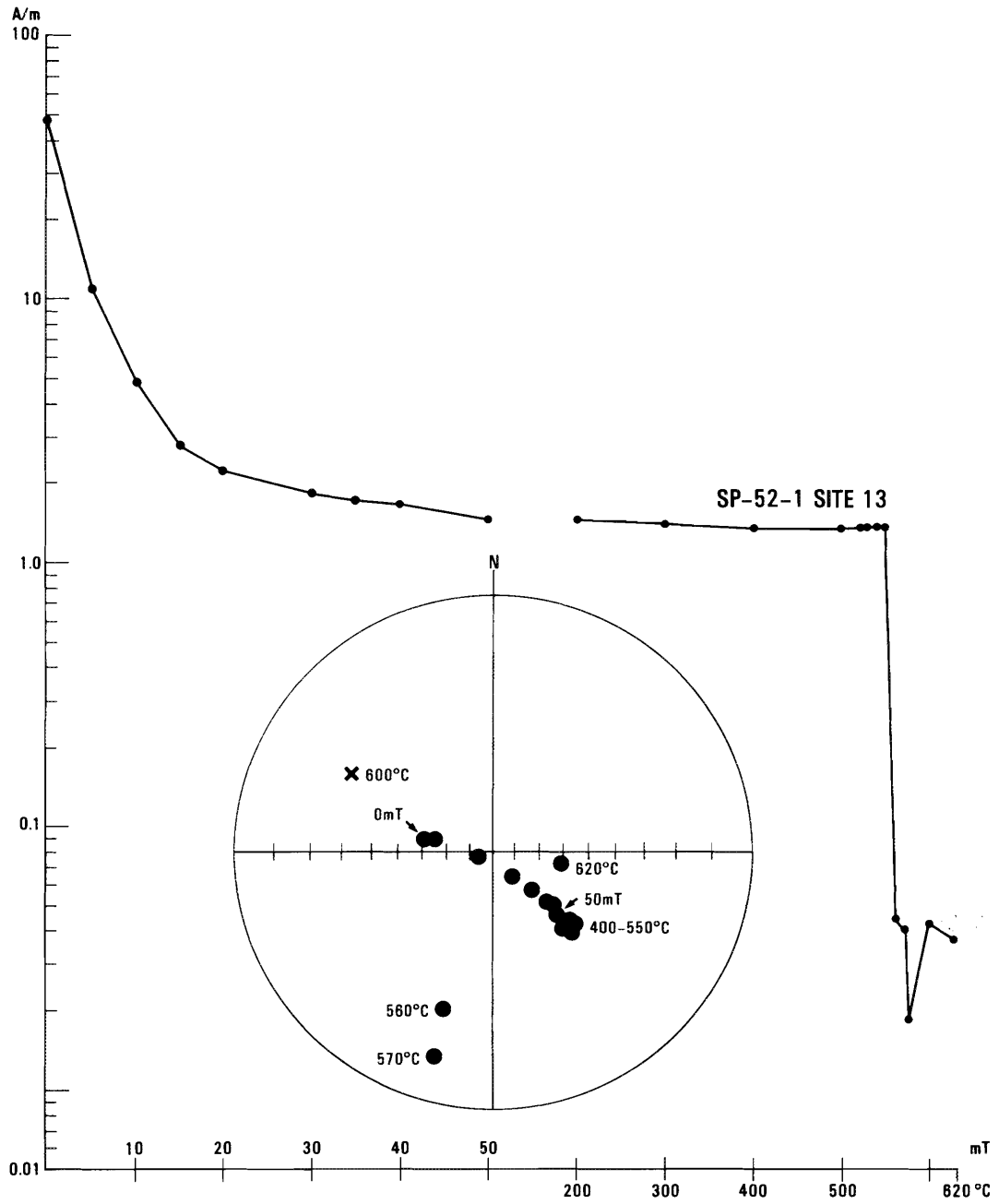


FIG. 4. Directional and intensity changes in specimen SP-52-1 from site 13. Thermal demagnetization follows AF demagnetization, which in this specimen was carried up to 50 mT. Thermal demagnetization in the 400–500°C range firmly established the end point, which was not quite reached by the AF treatment alone.

nent is stable under thermal demagnetization to temperatures of 525°C, after which there is a sharp drop in intensity. At the 550°C step viscous components begin to be troublesome, but the average direction from SP-68-1 and SP-69-1 at 550°C is close to being antiparallel to the A component, *i.e.*, reversed A. At higher temperatures, between and within sample (repeat measurements of remanence) consistency is lost.

In Fig. 9, the relative merits of AF and thermal demagnetization in isolating the A component are compared. On sample U-5-2 AF demagnetization was performed prior to thermal demagnetization, whereas in U-5-3 thermal demagnetization alone

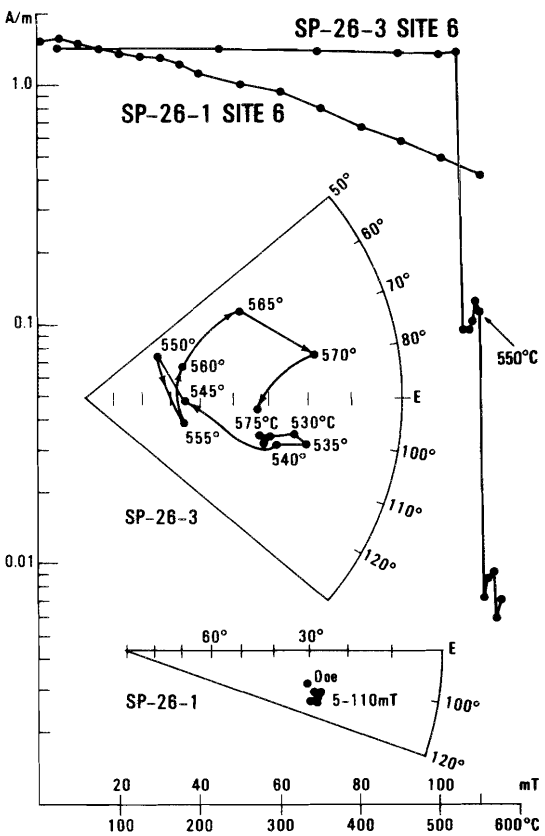


FIG. 5. Comparison of AF and thermal demagnetization results from two specimens from the same core. In the portion of the stereonet displaying the directional data for SP-26-3, the directional data are generalized in the 25–520°C range.

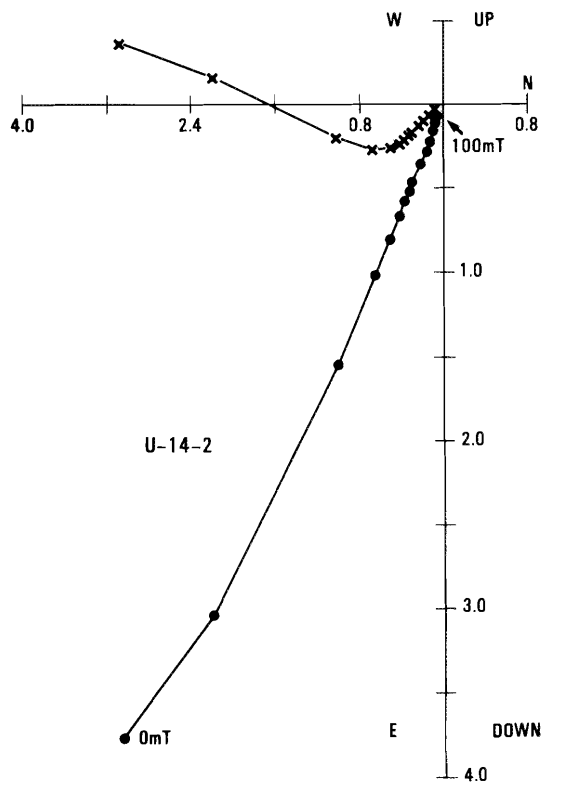


FIG. 6. Orthogonal demagnetization projection (Zijderveld 1967) of the successive end points of the magnetization vector in specimen U-14-2 during progressive AF demagnetization. Solid circles represent projections on the horizontal plane, whereas ×'s are those on the north–south vertical plane. Units indicated for the axes are in Am^{-1} (10^{-3} emu/cm^3).

was employed. U-5-2 exhibits similar behaviour to the samples from site 17a except that the A component unblocks at a somewhat lower temperature (500°C), and the reversed A component is better defined at the 510°C, 520°C, and 530°C steps. Where thermal demagnetization alone is used in U-5-3, the A component is not isolated. A very short trend of migration toward the A component is established at heatings up to 510°C, but the A component itself is not isolated. With subsequent heatings at 535°C, 540°C, and 545°C, directions nearly antiparallel to the A component are obtained. Similar behaviour, though not illustrated, is found in block sample U-3; the A component can only be isolated by AF demagnetization, but when isolated it is unblocked at 530°C and the reversed A component is isolated at higher temperatures, in the 540–555°C range.

Carriers of the NRM

The AF demagnetization experiments indicate that the characteristic A and B remanences in the stably magnetized samples of the Umfraville Gabbro possess high coercivities exceeding 100 mT. This fact points to either hematite or fine-particle magnetite as being the remanence carrier. Petrographic examination of thin sections and polished thin sections indicates the general absence of hematite in these rocks. Although large homogeneous magnetite and ilmenite are commonly present in most samples and hemoilmenite is present in a few, a less conspicuous but omnipresent feature in the stably magnetized samples is the presence of a dense clouding of the pyroxenes. Under oil immersion in reflected light these oxide inclusions appear similar to those illustrated by Evans *et al.* (1968). The largest of these inclusions have the optical properties of magnetite. Rodlets of an oxide? phase, smaller still than those in the pyroxene, appear as inclusions in plagioclase. These are similar in form and in dimensions to those illustrated by Haggerty (1976) in his figure Hg-36a.

Curie-point determinations also suggest that magnetite is the only ferrimagnetic phase in these rocks. Hemoilmenite may be a contributor inasmuch as the titanohematite in naturally occurring

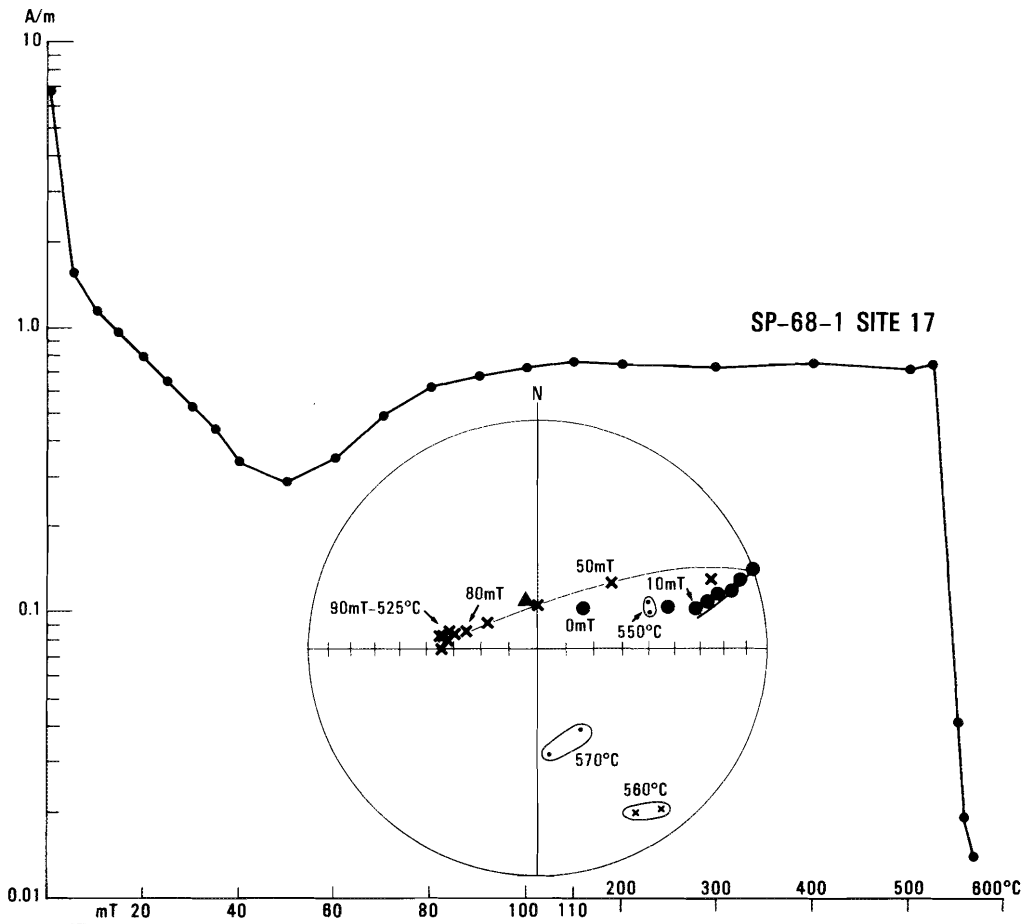


FIG. 7. Directional and intensity changes in SP-68-1 from site 17. The great circle path shown is that obtained from the 10 mT to 110 mT data using the method of Halls (1976).

hemoilmenite contains in excess of 20 mol percent ilmenite (Carmichael 1961; Kretchsmar and McNutt 1971; Lindsley 1973) and its high field Curie temperature will be comparable to that of low Ti-magnetite. $J_{sat.}$ *vs.* T curves were plotted for several samples from each of the paleomagnetic types: A and B remanence types and from a paleomagnetically unstable sample. All samples so examined exhibited simple curves having no low temperature inflection points and yielding single Curie points in the narrow temperature range of 565–585°C. Heating and cooling curves are similar in form, with the cooling curve generally lying slightly below the heating curve, indicating a small amount of oxidation during the experiment. Zero field Curie temperatures, *i.e.*, unblocking temperatures of the natural remanence, obtained during the thermal demagnetization experiments are in the range: 525–560°C (Fig. 10), which is also consistent with magnetite being the carrier of the stable component of NRM.

Discussion

Paleomagnetic data from rocks of the Grenville Province, when plotted in terms of pole positions, form a linear array extending from Australia to the equatorial central Pacific (Irving and McGlynn 1976; Buchan and Dunlop 1976), or in terms of equivalent antipoles, from near Newfoundland to the equatorial central Atlantic (Ueno *et al.* 1975). This array of poles has been termed the Grenville Track (Irving *et al.* 1974). The central problem with the Grenville Track is establishing its age in absolute terms and in establishing the relative age of poles that form the track, *i.e.*, the sense of 'younging'. Because of the deep-seated, plutonic nature of the rocks involved, superposition and cross-cutting relationships of classical stratigraphy are generally not available. The sense of younging of the Grenville Polar Track (G.P.T.) has been inferred largely on the basis of laboratory observed blocking temperatures in rock units which have yielded two or more poles spread out along the G.P.T. (Irving *et*

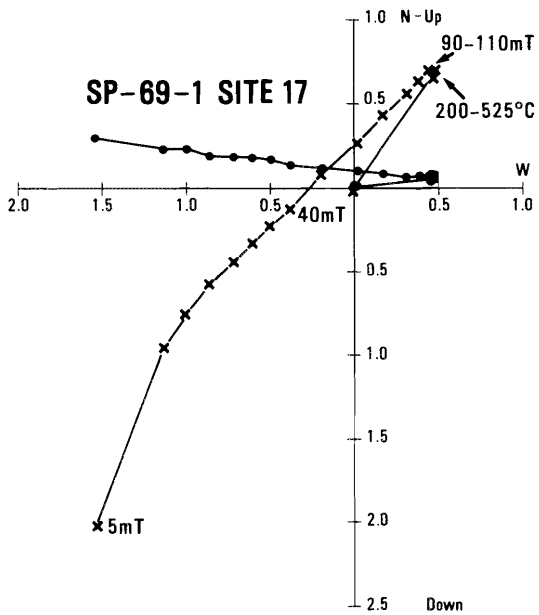


FIG. 8. Zijderveld plot of the components of magnetization during AF demagnetization followed by thermal demagnetization in SP-69-1 from site 17. The symbol conventions are as in Fig. 6. To preserve some detail in the diagram, the NRM data points are omitted and the plot begins with the 5 mT result.

al. 1974; Buchan and Dunlop 1976; Ueno *et al.* 1975). In the above-mentioned studies, the model yields a sense of younging from south to north (Australia to the equator) in terms of Pacific hemisphere poles. The assumption involved is that the remanent magnetizations are thermal remanent magnetization (TRM's) acquired during progressive cooling, during which time polar motion occurred. Thus in rock units where two or more magnetizations are isolated, lower blocking temperature remanences correspond to younger paleomagnetic poles. The assumption that the magnetizations are TRM's is probably a good one, at least in a number of studies, where a single component is shown to be carried by both hematite and magnetite (Hargraves and Burt 1967; Buchan and Dunlop 1976; McWilliams and Dunlop 1975). However, Buchan (1978) has recently shown in a study of the Thanet complex, the northern exposures of which are 8 km south of the Umfraville Gabbro body, that the cooling history or the mechanisms of magnetization have to be more complex. If the three components of magnetization isolated in the Thanet complex were assumed to have been acquired by TRM mechanisms and the thermal history were one of simple progressive cooling, then the polar motion inferred would be from north to south, which is the opposite sense of younging inferred in the earlier studies of Grenville rocks by Irving *et al.* (1974); Buchan and Dunlop (1976); and Ueno *et al.* (1975).

In the present study, however, the sense of younging of the Grenville Track cannot be inferred from the unblocking temperatures (Fig. 10). Unblocking temperatures are in a narrow range, 525–560°C, but within this range both high and low values occur at either end of the Grenville Track. Since the exsolution of oxide particles in silicates is a diffusional process, temperature can be traded for time. Anderson (1966) has shown that the ilmenohematite rodlets in feldspar of the Labrieville Anorthosite, Quebec, exsolved at temperatures in excess of 850°C. On the other hand, magnetite rodlets in pyroxene have exsolved at substantially lower temperature conditions of the amphibolite to granulite facies of regional metamorphism in the Grenville Front region (Palmer *et al.* 1977). The explanation we offer is that in some of the Umfraville samples the magnetization was not acquired by a TRM mechanism, but is rather due to the nucleation and growth of fine magnetite particles (now appearing as exsolved platelets (?) and rods in silicates) at temperatures that are high (~500°C), but still less than the Curie temperature. This probably occurred when the rate of cooling of the intrusive body slowed as the intrusive temperatures approached the ambient temperatures of the enclosing amphibolite-facies country rocks.

The establishment of the absolute age of the poles forming the Grenville Track and even the establishment of the age limits of the track itself is a subject of considerable conjecture. This stems from the fact that rocks as old as Archean are involved in the Grenville Province and that several periods of igneous activity and metamorphism during the Proterozoic are recognized (Krogh and Davis 1971; Doig 1977). Several models have been proposed for the age of the Grenville Track.

Model one assumes a rather old age for Grenville poles corresponding to Rb–Sr whole-rock isochron ages of 1200 Ma or more (Roy and Fahrig 1973; Fahrig and Jones 1976). Model two assumes an 1150–1050 Ma age for the Grenville Track, and hence essential contemporaneity with the middle to upper Keweenawan track defined principally from rocks in the Lake Superior region—well removed from the Grenville Province. This age assignment has plate tectonic implications of a convergence between a separate continent "Grenvillia" with proto-North America—"interior Laurentia". In this second model collision or 'docking' occurred at about 1000 Ma when the Grenville Track and the Keweenawan Track converge in the equatorial Pacific. In elementary form this model was advocated by Irving *et al.* (1972) and Palmer and Car-

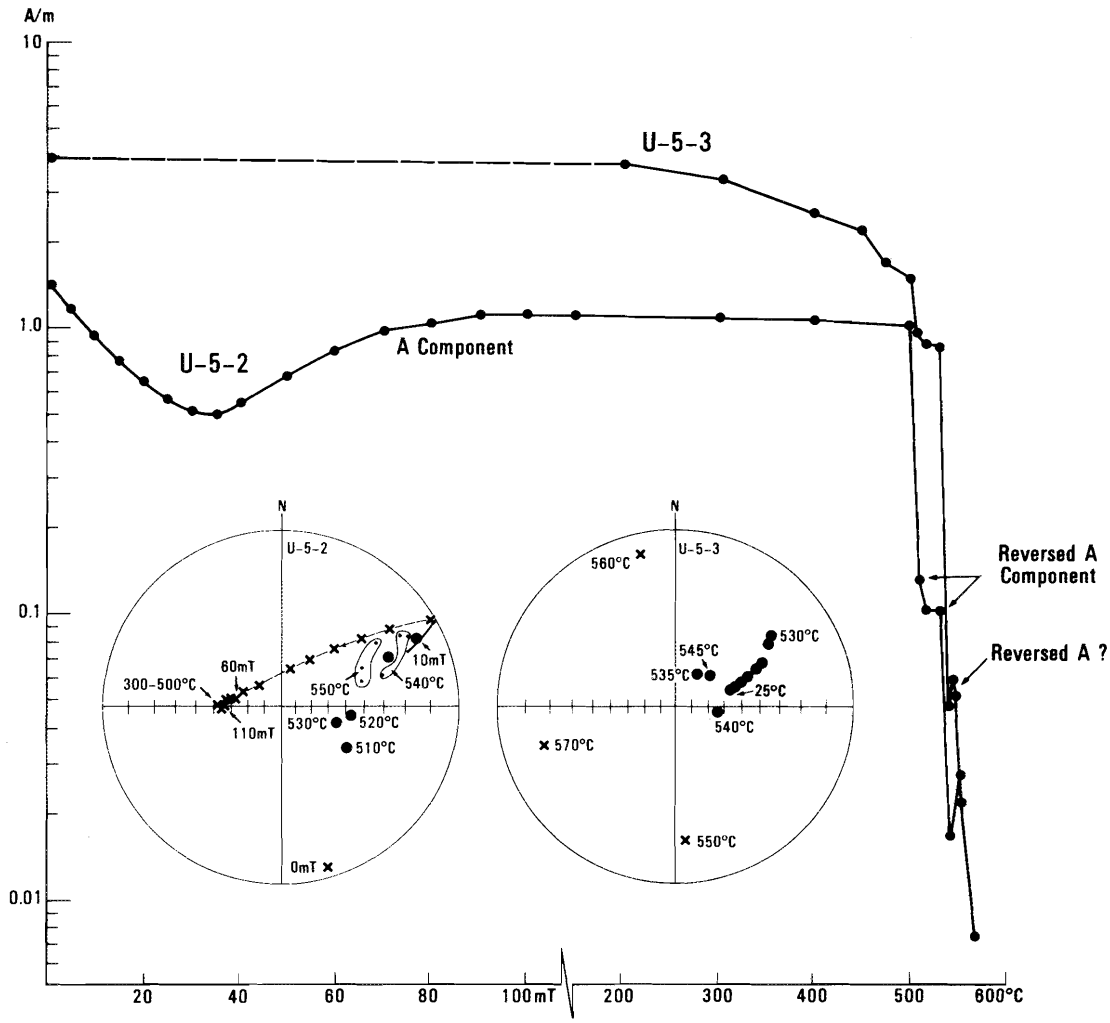


FIG. 9. In specimen U-5-3 thermal demagnetization only is used; in U-5-2 thermal demagnetization follows AF demagnetization to a field of 110 mT. The A component is isolated by AF treatment, but not by thermal treatment. The reversed A component is isolated in the 510–530°C range (U-5-2) and in the 535–545°C range (U-5-3). In the stereonet plots, closed circles (×'s) are north-seeking directions on the lower (upper) hemisphere.

michael (1972). More refined discussions of this model are to be found in Irving *et al.* (1974) and Irving and McGlynn (1976). A third model assumes a still younger age for the Grenville Track, corresponding to Rb–Sr mineral isochrons and K–Ar whole-rock and mineral ages of 1050–900 Ma or less (Du Bois 1962; Hargraves and fish 1972; McWilliams and Dunlop 1975; Patchett and Bylund 1977; Morris and Roy 1977). A weakness of both models one and three is that until recently both were defined by Grenville or equivalent Sveco-Norwegian poles exclusively.

However, several recent paleomagnetic studies of late Proterozoic units within 'interior Laurentia' lend some support to model three. Elston and Gromme (1974) reported a pole from the upper member of the Nankoweap Formation of Arizona which falls on the periphery of the Grenville Track (Fig. 11). The Nankoweap Formation rests unconformably (Elston and Scott 1976) on the 1090 Ma Cardenas Lavas (McKee and Noble 1976). Many of the formations underlying the Nankoweap Formation yield poles similar to those of the Keweenawan Track (Elston and Gromme 1974) and therefore the Nankoweap pole is likely younger than the 1050 Ma Keweenawan Freda and Nonesuch poles (Henry *et al.* 1977). Furthermore Roy and Robertson (1978) have demonstrated that

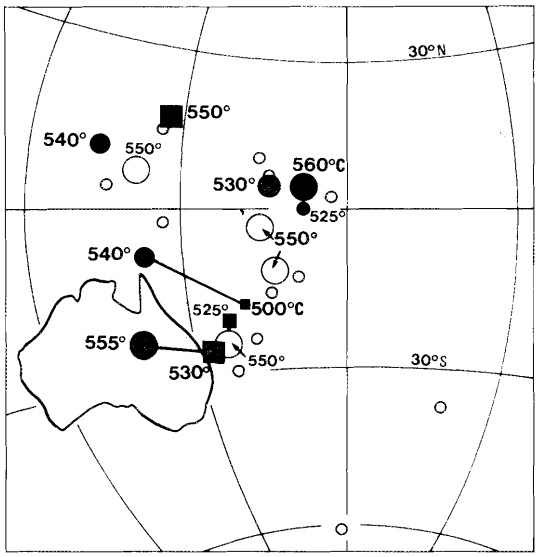


FIG. 10. Site poles as listed in Table 2 with A type poles as squares and B type poles as circles. Blocking temperatures are indicated for those sites where data are available, with symbol size proportional to T_B. In those sites where the A direction is succeeded at higher temperatures by reversed A (or B) directions, the two corresponding poles are joined by a solid line.

the Jacobsville poles continue the late Keweenawan trend of poles across the equator to 16° south latitude. Because of a longitude difference these Jacobsville poles do not lie along the Grenville Track, but the argument is made that they form the complementary portion of a Grenville Loop in which the Grenville Track is the second limb.

Considered in isolation the available geochronology on the Umfraville Gabbro is compatible with either model two or model three. We tentatively accept that the zircon data provide the best estimate of the time of intrusion of the pluton and that our 911 Ma K–Ar isochron reflects the time at which the body had cooled to temperatures of 300°C or so. Whether the magnetization was acquired by a thermo-chemical remanent magnetization (TCRM) mechanism or by a TRM mechanism, it would have been acquired at temperatures less than the observed blocking temperatures of 525–560°C. Those remanences that unblock in the 525–530°C range may well have been acquired at temperatures of 450°C if cooling times of the order of 10^6 years are involved (Pullaiah *et al.* 1975). Cooling times of this length are not unreasonable given the regional amphibolite facies of the surrounding terrain. Although the arguments are not definitive we consider it more likely that the magnetization in the Umfraville Gabbro was acquired closer to 911 Ma when the K–Ar system closed

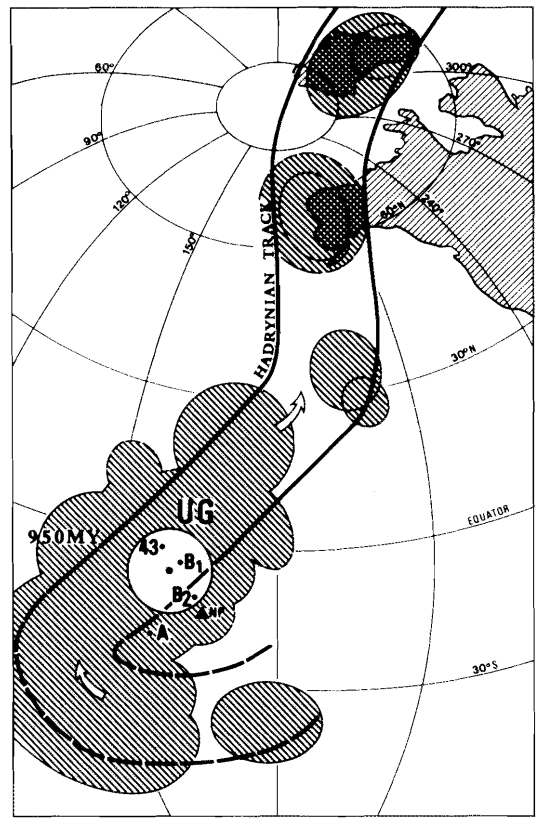


FIG. 11. The Umfraville pole (UG), together with the A and B component poles of this study, plotted separately as A and B_1. B_2 and 43 are the single characteristic remanence poles isolated by Symons (1978) and Hood (1958) respectively. These poles are shown in the context of other Hadrynian data as summarized by Roy and Robertson (1978). Poles south of the equator are derived from Grenville Province rock units with the single exception of the Nankoweap pole (NP) from the Grand Canyon Supergroup of Arizona.

(Fig. 11) rather than at the time when the body was intruded and crystallized in the crust at about 1160 Ma.

Our overall mean direction and our B component direction (Fig. 11) is statistically indistinguishable from the result recently published by Symons (1978). Symons, however, did not find clear-cut evidence for the A component, which is not surprising inasmuch as the A component seems to be restricted to rocks in a limited geographic region not included in his sampling traverses. We differ from Symons (1978) only in the age assignment of the magnetization; our assignment makes the magnetic history compatible with our model three and his model four (after McWilliams and Dunlop).

Acknowledgments

After this paper was submitted but prior to its

acceptance we received pre-prints of papers subsequently published by Symons (1978) and Buchan (1978), which helped us in the revision process. We are grateful to S. B. Lumbers for advice and assistance in the sampling of the Umfraville Gabbro. The manuscript has benefited from critical review by C. M. Carmichael. This work was supported by National Research Council Grants A3544 to H. C. Palmer and A8163 to A. Hayatsu.

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